Wildfire Detection and Visualization using ML-Aided Tools and Sensor Fusion

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Machine Learning and Autonomy for Diverse Domains (MADD Lab)

Introduction

Current wildfire monitoring technologies rely heavily on resource-intensive systems, including satellite thermal imagery, ground-based thermal sensors, and observation towers. While these methods provide valuable data for interpreting wildfire behavior, they often fall short in terms of timely detection and are frequently used only after a fire has already occurred. My research proposes the use of **autonomous UAVs equipped with diverse sensors** for real-time fire detection.

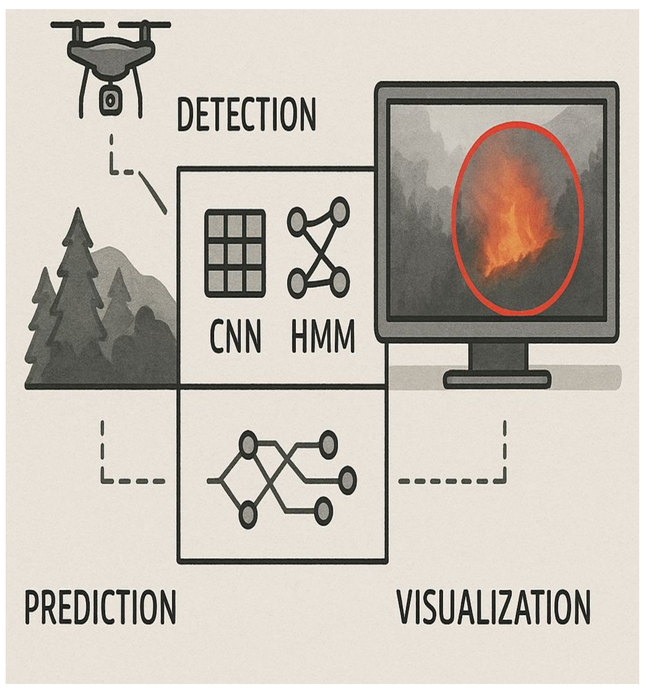


Figure 1: Sketch of the current system

Contributions:

- Design an in-house drone swarm system with low-cost thermal and visual sensors.
- Extend perception capabilities of a single drone to a swarm system.
- Deploy deep learning-based fire detection models on a Jetson Orin Nano platform for online detection.

Hardware

- **Holybro Drone**
  - Pixhawk 6C flight controller, an onboard GPS, and telemetry radio.
  - Cost ~ $735



Figure 2: Holybro Drone

- **Jetson Orin Nano**
  - AI supercomputer
  - 1024 NVIDIA Cuda Cores
  - Cost ~ $250



Figure 3: Jetson Orin Nano

- **RealSense Intel depth camera**
  - RGB camera & IR projector
  - 1280 x 800 at 30 fps
  - Cost ~ $314



Figure 4: RealSense Camera

- **FLIR Lepton 3.5**
  - Thermal module
  - Ranges from -10℃ to 110℃
  - Cost ~ $199

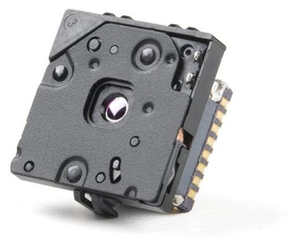


Figure 5: FLIR thermal module

Testing

Sample Visual from Flight Tests

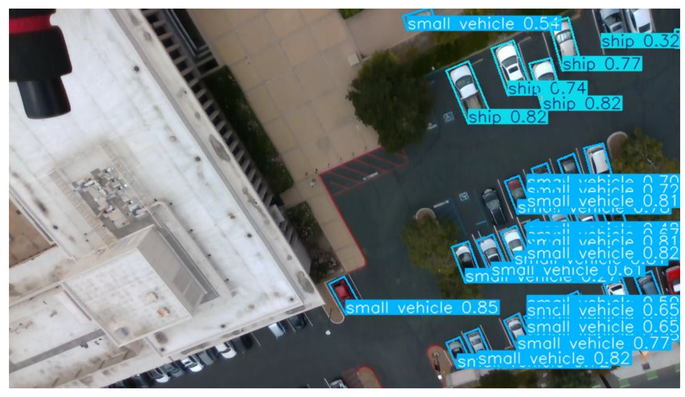


Figure 2: Holybro Drone

Above is a frame captured by the Intel RealSense depth camera. It shows the output of a pre-trained object detection algorithm. In this frame, the algorithm achieved twenty correct detections and five missed detections.

Sample Visual From Thermal Testing

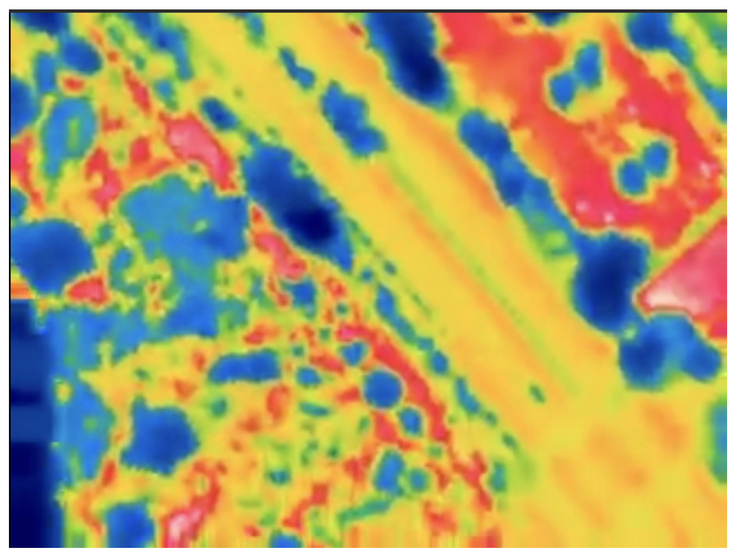


Figure 7: Holybro Drone

Figure 7 shows a thermal image captured from 300 feet above Foothill Blvd. The heat signatures highlight the contrast between the warmer road surface and cooler surrounding objects such as trees.

In the Fall, we plan to continue more exhaustive testing and build a scalable robust system of two or more drones. The proceeding drones will be equipped with a Raspberry Pi and different sensors for more fire data.